

Problem

Develop the state equations for the problem you designed in Prob. 16.13.

Step-by-step solution

Step 1 of 1

Develop the state equations for the problem you designed in Prob. 16.2.

Although there is no correct way to work this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

Develop the state equations for Problem 16.2.

Chapter 16, Problem 2.

Find v_x in the circuit shown in Fig. 16.39 given $v_s = 4u(t)$ V.

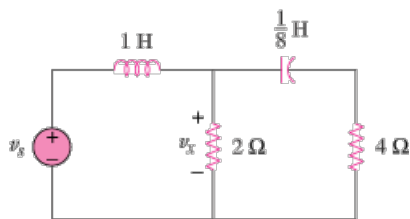
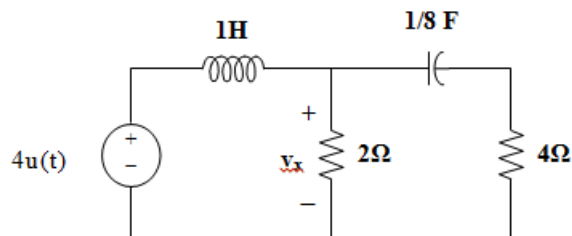


Figure 16.39

Solution

First select the inductor current i_L and the capacitor voltage v_C to be the state variables.

Applying KCL we get:

$$-i_L + \frac{v_x}{2} + \frac{v_C}{8} = 0; \text{ or } v_C = 8i_L - 4v_x$$

$$i_L' = 4u(t) - v_x$$

$$v_x = v_C + 4 \frac{v_C}{8} = v_C + \frac{v_C}{2} = v_C + 4i_L - 2v_x; \text{ or } v_x = 0.3333v_C + 1.3333i_L$$

$$v_C' = 8i_L - 1.3333v_C - 5.3333i_L = -1.3333v_C + 2.666i_L$$

$$i_L' = 4u(t) - 0.3333v_C - 1.3333i_L$$

Now we can write the state equations.

$$\begin{bmatrix} v_C' \\ i_L' \end{bmatrix} = \begin{bmatrix} -1.3333 & 2.666 \\ -0.3333 & -1.3333 \end{bmatrix} \begin{bmatrix} v_C \\ i_L \end{bmatrix} + \begin{bmatrix} 0 \\ 4 \end{bmatrix} u(t); \quad v_x = \begin{bmatrix} 0.3333 & 1.3333 \end{bmatrix} \begin{bmatrix} v_C \\ i_L \end{bmatrix}$$